Sepid (Sepidehsadat) Hosseini

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Education

Simon Fraser University — British Columbia, Canada
PhD in Computing Science, GPA: 4.11/4.3 — May 2025

- Field: 3D Computer Vision and Computer Graphics.

Seoul National University — Seoul, South Korea
Master of Science in Electrical and Computer Engineering, GPA: 95.1/100 — August 2019

- Field: Intelligent Signal Processing, Computer Vision, Facial and Gesture Recognition.

Seoul National University — Seoul, South Korea
Bachelor of Science in Electrical and Computer Engineering, GPA: 88.6/100 — March 2017

- Field: Smart Road Implementation Using Arduino Car.

Work Experience

RBC Borealis (Borealis AI) — Vancouver, Canada
Senior Machine Learning Researcher / Tech Lead — May 2026–Present

- Technical lead for ATOM, the Asynchronous Temporal Model, a foundation model for financial services focused on fraud detection, anomaly detection, and personalization.
- Leading research-to-production delivery of large-scale temporal modeling solutions in collaboration with research, engineering, and product teams.

RBC Borealis (Borealis AI) — Vancouver, Canada
Machine Learning Researcher — Dec 2023–May 2026

- Core researcher on the fraud detection team, contributing to ATOM and its adaptation to recommender systems and personalized financial applications.
- Conducted applied machine learning research leading to publications, patents, and production-oriented model development.

RBC Borealis (Borealis AI) — Vancouver, Canada
Machine Learning Research Intern — May 2023–Sep 2023

- Conducted research on prompt-based learning for efficient temporal domain generalization under distribution shift.

LG Electronics, CTO, Intelligent Systems Research Center — Seoul, Korea
Research Intern — Jul 2017–Sep 2017

- Developed capsule-network-based methods for face spoofing detection and robust facial analysis.

Samsung Techwin (Hanwha) — Seoul, Korea
Software Intern — Jul 2015–Aug 2015

- Worked on object tracking algorithms for video analysis and surveillance-related applications.

Selected Publications

FairNVT: Fair Classification via Noise Injection in Vision Transformers — TMLR 2026
Qiaoyue Tang, **Sepidehsadat Hosseini**, Mengyao Zhai, Thibaut Durand, Greg Mori

Scalable Temporal Domain Generalization via Prompting — PUT Workshop, ICML 2025
Sepidehsadat Hosseini, Mengyao Zhai, Hossein Hajimirsadeghi, Frederick Tung

Radar: Fast Long-Context Decoding for Any Transformer — ICLR 2025
Yongchang Hao, Mengyao Zhai, Hossein Hajimirsadeghi, **Sepidehsadat Hosseini**, Fred Tung

PuzzleFusion: Unleashing the Power of Diffusion Models for Spatial Puzzle Solving — NeurIPS 2023 Spotlight
Sepidehsadat Hosseini, Mohammad Amin Shabani, Saghar Irandoust, Yasutaka Furukawa

Floorplan Restoration by Structure Hallucinating Transformer Cascades — BMVC 2023
Sepidehsadat Hosseini, Yasutaka Furukawa

HouseDiffusion: Vector Floorplan Generation via a Diffusion Model with Discrete and Continuous Denoising — CVPR 2023
Mohammad Amin Shabani, **Sepidehsadat Hosseini**, Yasutaka Furukawa

House-GAN++: Generative Adversarial Layout Refinement Network — CVPR 2021
Nelson Nauata, **Sepidehsadat Hosseini**, Kai-Hung Chang, Hang Chu, Chin-Yi Cheng, Yasutaka Furukawa

GF-CapsNet: Using Gabor Jet and Capsule Networks for Facial Age, Gender, and Expression Recognition — FG 2019
Sepidehsadat Hosseini, Nam Ik Cho

Patents

Neural Network Temporal Domain Generalization Method and System — US App. 18/891,940
Sepidehsadat Hosseini, Mengyao Zhai, Hossein Hajimirsadeghi, Frederick Tung. Assignee: Royal Bank of Canada. Filed 2024.

Application of Representation Learning to Discrete Event Sequences Across Discrete Domains — PCT/CA2025/050052
Mohamed Osama Ahmed, Taivanbat Badamdorj, Ruizhi Deng, Peter Forsyth, **Sepidehsadat Hosseini**, Greg Mori, Gabriel L. Oliveira, Edward Smith, Frederick Tung, Ankit Vani, He Zhao. Assignee: Royal Bank of Canada. Filed 2025.

Memberships and Volunteering Experience

Catch, Adapt, and Operate: Monitoring ML Models Under Drift — Rio de Janeiro, Brazil
Lead Organizer and Program Chair — ICLR 2026
- Led the organization and program committee for an ICLR workshop on monitoring, adapting, and operating machine learning models under distribution shift.

LG Global Challenger — Seoul, Korea
Participant, Student Ambassador — 2016–2017
- Selected for a fellowship program to travel across South Korea and promote Korean culture and innovation.

Seoul National University Dormitory — Seoul, Korea
Student Advisor — 2013–2014
- Supported international and domestic students in residential life and community-building activities.

Bean Seoul — Seoul, Korea
Volunteer Instructor — 2013–2017
- Volunteered in educational and recreational activities with children in orphanages.

IUST Robotics Team — Tehran, Iran
Team Member — 2009–2012
- Participated in robotics competitions as part of the university robotics research team.

Honors and Awards

SFU Computing Science Graduate Fellowship
Fall 2019, Fall 2020, Fall 2021, Summer 2022, Fall 2022

SFU Computing Science Entrance Scholarship
Fall 2019
- Award valued at CAD 10,000.

Seoul National University Excellent Student Global Scholarship
2017, 2018
- Award valued at USD 8,000 per year.

Brain Korea Scholarship, Excellent Researcher Award
2017, 2018

LG International Student Ambassador
2016–2017

Korean Government Scholarship
2012–2017
- Award valued at approximately USD 97,000.